

CENTREVILLE, AL, USA

WMO: 722290

Lat: 32.900N

Lon: 87.250W

Elev: 138

StdP: 99.68

Time Zone: -6.00 (NAC)

Period: 86-94

WBAN: 03881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.2	(c) -2.8	(d) -13.2	(e) 1.2	(f) -0.6	(g) -10.7	(h) 1.5	(i) 1.9	(j) 7.7	(k) 9.6	(l) 7.1	(m) 8.6	(n) 3.1	(o) 0	(p) 0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	34.4	24.3	33.3	24.2	32.3	23.9	25.8	31.7	25.3	31.1	24.9	30.3	2.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.5	28.6	23.6	18.8	27.9	23.2	18.3	27.4	80.4	31.5	77.9	31.2	75.9	30.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.9	5.9	5.5	-9.9	36.5	3.4	0.6	-12.3	36.9	-14.3	37.3	-16.2	37.6	-18.7	38.0
			-10.8	27.2	3.1	0.7	-13.1	27.7	-14.9	28.1	-16.7	28.4	-18.9	28.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	7.5	10.0	13.5	17.1	21.3	25.1	26.7	26.4	23.7	17.1	12.9	8.5
	DBStd	7.84	4.64	5.35	4.96	4.19	2.91	2.00	1.55	2.16	3.15	4.06	4.91	5.46
	HDD10.0	311	102	59	23	4	0	0	0	0	0	3	28	92
	HDD18.3	1358	334	234	161	68	9	0	0	1	4	71	171	306
	CDD10.0	3056	26	60	131	218	349	452	518	507	410	223	114	47
	CDD18.3	1061	0	2	9	33	100	202	260	249	164	33	7	2
	CDH23.3	9291	0	7	96	328	757	1808	2408	2320	1289	247	28	2
	CDH26.7	3706	0	0	7	64	198	757	1104	1048	476	52	1	0
Wind	WSAvg	2.8	3.1	3.3	3.4	3.1	2.7	2.4	2.2	2.3	2.5	2.8	3.1	3.2
Precipitation	PrecAvg	1372	135	140	140	113	87	110	121	98	101	72	112	115
	PrecMax	1606	255	271	248	215	219	229	272	157	210	222	218	194
	PrecMin	1033	63	42	36	42	4	32	26	31	23	1	41	72
	PrecStd	146	57	57	55	47	56	57	63	36	50	48	49	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	24.4	27.7	30.5	31.2	34.6	36.1	35.9	33.9	30.7	25.8	23.1
		MCWB	15.3	18.7	17.4	19.2	21.5	23.1	25.1	24.4	23.7	21.9	18.4	19.7
	2%	DB	18.9	21.9	25.8	28.4	30.1	33.1	34.5	34.3	32.3	28.0	24.0	20.6
		MCWB	14.1	16.0	16.1	18.5	21.3	23.1	24.7	24.4	23.4	19.9	17.6	17.7
	5%	DB	16.9	20.0	24.2	26.9	29.0	32.0	33.3	33.1	31.0	26.1	22.4	18.8
		MCWB	13.1	15.2	15.9	18.1	21.0	23.1	24.7	24.2	23.1	18.7	17.0	15.4
	10%	DB	15.2	18.4	22.2	25.2	27.6	30.8	32.1	31.9	29.6	24.2	20.8	17.0
		MCWB	12.1	14.5	15.1	17.1	20.4	22.9	24.4	24.0	22.8	17.6	16.8	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	19.5	20.6	21.9	24.0	25.7	26.9	26.5	25.7	23.5	21.4	20.3
		MCDB	19.6	22.9	23.2	25.9	28.8	31.0	33.0	32.2	30.4	28.5	23.7	22.2
	2%	WB	15.9	18.2	18.7	20.9	23.2	24.9	26.1	25.7	24.9	22.0	20.1	18.5
		MCDB	18.0	20.5	22.6	25.1	27.6	30.3	32.3	31.7	30.0	25.1	21.9	20.2
	5%	WB	14.4	17.0	17.5	19.7	22.4	24.4	25.5	25.2	24.3	20.7	18.9	16.6
		MCDB	16.1	19.1	22.0	24.6	26.5	29.9	31.4	31.1	29.1	24.0	21.0	18.2
	10%	WB	12.3	15.2	16.2	18.6	21.6	23.8	25.0	24.6	23.7	19.4	17.6	14.8
		MCDB	14.2	17.5	20.2	23.9	25.6	29.2	30.5	30.1	28.1	22.7	19.8	16.5
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.0	12.1	12.7	10.8	10.9	10.5	10.6	10.4	12.0	11.3	10.0
		MCDBR	12.8	12.1	14.4	14.8	12.1	12.5	12.2	12.2	11.5	13.1	12.1	10.9
		MCWBR	9.1	7.6	7.4	6.9	5.1	5.0	4.6	4.1	4.9	6.6	7.7	8.3
	5% WB	MCDBR	9.8	9.6	11.1	11.7	9.7	11.0	11.1	10.9	10.1	10.0	9.1	9.2
	MCWBR	8.9	7.3	7.3	6.7	4.8	5.1	4.7	4.2	4.8	5.7	7.7	8.4	
Clear-Sky Solar Irradiance	taub	0.319	0.329	0.358	0.399	0.451	0.482	0.540	0.536	0.443	0.368	0.342	0.321	
	taud	2.531	2.501	2.435	2.302	2.204	2.154	2.007	2.014	2.261	2.461	2.477	2.540	
	Ebn at Noon	902	927	920	891	843	813	764	761	823	870	865	879	
	Edh at Noon	84	95	110	131	145	152	176	172	129	98	88	79	
All-Sky Solar Radiation	RadAvg	2.70	3.25	4.39	5.55	6.12	6.26	6.02	5.56	4.92	4.11	3.06	2.31	
	RadStd	0.20	0.43	0.41	0.42	0.44	0.48	0.39	0.33	0.41	0.50	0.31	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+107

Nomenclature: See separate page